Alternating Optimization for LLM Post-Training: From Divergence to Verified Solutions

The Chinese University of Hong Kong — Department of Computer Science and Engineering

jianghuanyun

Supervisor: Prof. Hoi To Wai

ALS & its 3 repairs → DIVERGE | SGD / AdamW / LoRA → CONVERGE | proof below (matched budgets, pretrained baselines shown)

Research Question

Why does ALS-based post-training diverge on deep LLMs, and what works instead? ALS solves $W^* = (X^TX + \lambda I)^{-1}X^TY$ in closed form — one solve replacing hundreds of gradient steps. It converges on ≤24-layer models but diverges on every observed 28-layer attempt. Two sub-questions: (Q1) can the divergence be explained and fixed? (Q2) what actually works, compared fairly?

The 2×2 Framework — Four Post-Training Algorithms

Two design choices define how a model is fine-tuned: **which optimizer** drives the update, and **how many parameters** are trained. Crossing them gives four algorithms. Here is each one, in plain terms:

	Full-rank **train ALL weights**	**LoRA** **train a small** **low-rank adapter**
AdamW **adaptive-step** **gradient descent** **+ weight decay**	**B:** repeatedly compute each weight's slope (gradient) of the loss and step downhill; AdamW gives each weight an auto-tuned step size, so it descends quickly and stably. Updates every weight of the model. **1.25 PPL.**	**D:** freeze the whole model; train only two small matrices A (r×d) and B (d×r) whose product ΔW=BA (r ≪ d) approximates a full update with far fewer parameters. **10.41 PPL.**
ASP **alternating: ALS** **closed-form solve** **+ SGD + noise**	**A:** each cycle, ALS computes the least-squares-optimal output weights $W^*=(X^TX+\lambda I)^{-1}X^TY$ in one matrix solve — bypassing thousands of gradient steps — then SGD and noise follow. **DIVERGES on deep models.**	**C:** the same ALS + gradient + noise cycle, applied to the small adapter instead of full weights. Converges, but quality is an order of magnitude worse than AdamW. **135.4 PPL.**

All four run until they consume the **same total FLOPs** (one ALS solve ≫ one AdamW step), with shared data, seeds, and evaluation — so the comparison is fair.

The Story in 4 Steps

STEP 1 — ALS diverges on deep models

The closed-form solve $W^* = (X^TX + \lambda I)^{-1}X^TY$ converges on ≤24-layer models but diverges on every observed 28-layer attempt (11/11 on Qwen2.5-7B; PPL → NaN in 1–3 cycles).

STEP 2 — Mechanism proven by controlled ablation

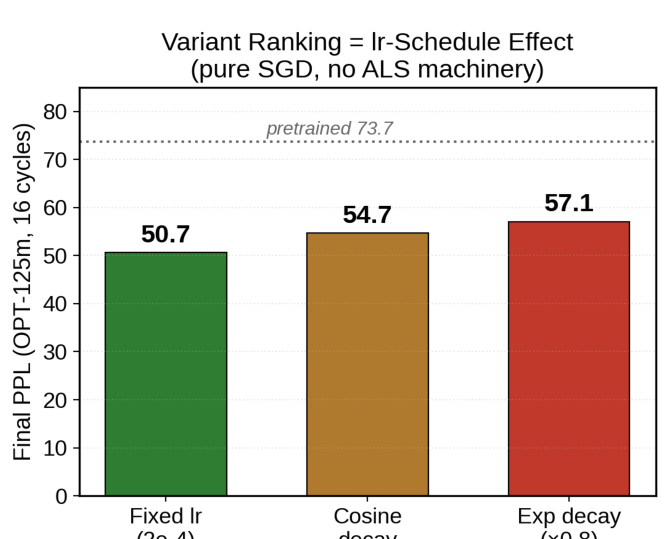
Five-condition ablation on Qwen2.5-7B isolates **ALS weight modification** as the necessary-and-sufficient cause within the tested space: every condition with a real weight solve diverges; SGD-only, perturbation-only and suppressed-weight ALS all converge (Fig. 5).

STEP 3 — All three natural repairs fail matched controls

Gradient injection, low-rank probing and soft-target distillation — each matched against an equal-budget SGD baseline — add no measurable value (Figs. 1–3).

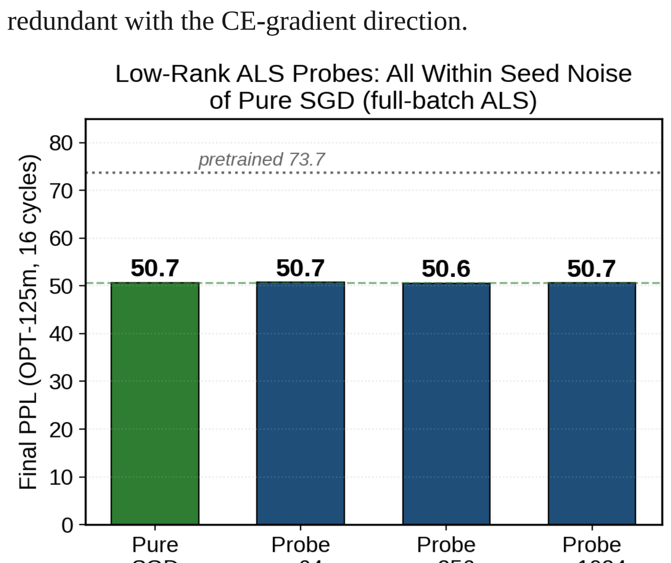
Repair analysis 1 — Variant ranking = learning-rate-schedule effect

With injection vacuous, the ranking is explained by the lr schedule alone (pure SGD, no ALS machinery): fixed 50.7 < cosine 54.7 < exp 57.1 — same order as the 7B variants.



Repair analysis 2 — Low-rank ALS probes: within seed noise at every rank

r = 64 / 256 / 1024 with full-batch ALS solves: 50.7 / 50.6 / 50.7 vs pure SGD 50.7 — no rank improves on SGD. The closed-form solve is redundant with the CE-gradient direction.



STEP 4 — Verified alternatives, baselines shown

AdamW full-rank 1.25 PPL; AdamW+LoRA 10.41 PPL at 0.1% trainable params; plain SGD 6.83 PPL. All far below the pretrained 73.1 PPL reference — **up to 58× improvement over doing nothing** (Fig. 4).

Contributions

- **Diagnosis** — first controlled evidence that ALS weight modification (not noise / optimizer / hook overhead) causes deep-model divergence, via a 5-condition ablation.
- **Theory** — residual-amplification model (ρ≈1.08/layer) predicts a 25–28-layer boundary, consistent with 8 architectures.
- **Saved design space** — three plausible ALS repairs proven ineffective by matched-budget controls; identifies why (closed-form solve ≡ CE gradient).
- **Verified alternatives** — AdamW 58×, SGD 10.7×, LoRA 7× below the 73.1 pretrained baseline, with a corrected evaluation harness.

Why ALS diverges — causal theory

Residual connections amplify an ALS perturbation by an **effective ρ ≈ 1.08 per layer** (phenomenological fit), i.e. ≈8× over 27 layers — beyond SGD's per-cycle recovery. The predicted boundary falls **between 25 and 28 layers**, consistent with the observed 24-converges / 28-diverges split.

The perturbation *grows* across cycles (δ: 0.085 → 0.196) — a positive feedback loop, which is why damping/clipping patches only delay the failure.

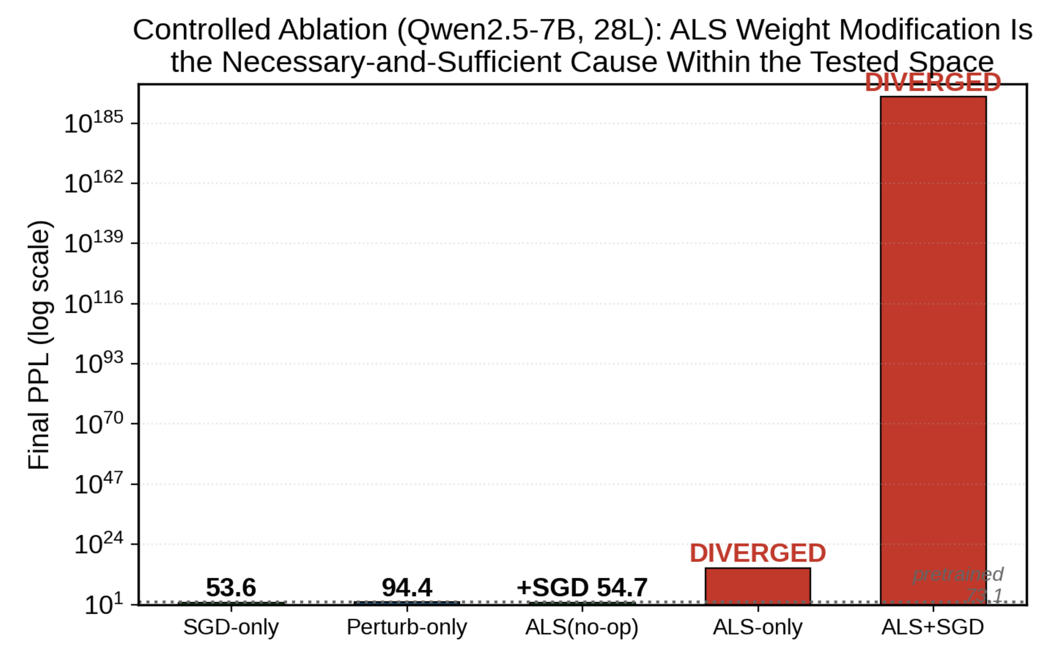
Method & discipline

- **2×2 factorial** (ASP vs AdamW × full-rank vs LoRA), FLOPs-matched, shared seeds/data.
- **Harness audit**: OPT numbers re-verified with correct tokenizer + pad-masked labels (baseline 73.7 PPL); 7B baseline 73.1 PPL.
- **Falsification discipline**: pre-stated hypothesis, open code, negative results reported with matched controls.

Key Findings & Experimental Results

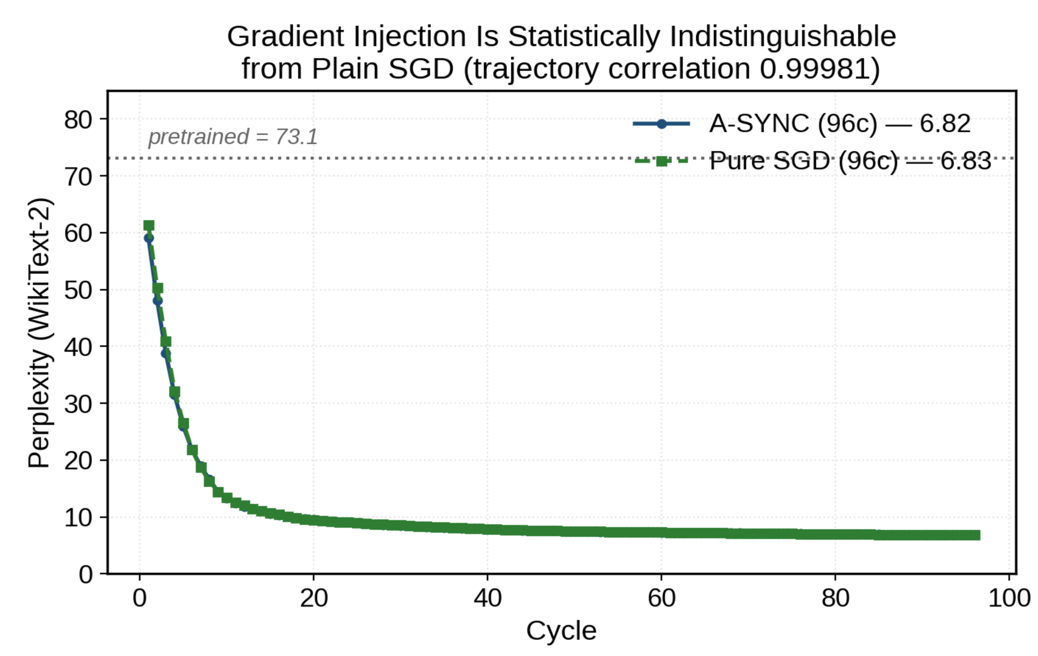
Finding 1 — ALS weight modification is the sole cause of divergence

A five-condition ablation on Qwen2.5-7B isolates the cause: every condition with a real weight solve diverges; SGD-only, perturbation-only and suppressed-weight ALS all converge on the same model.



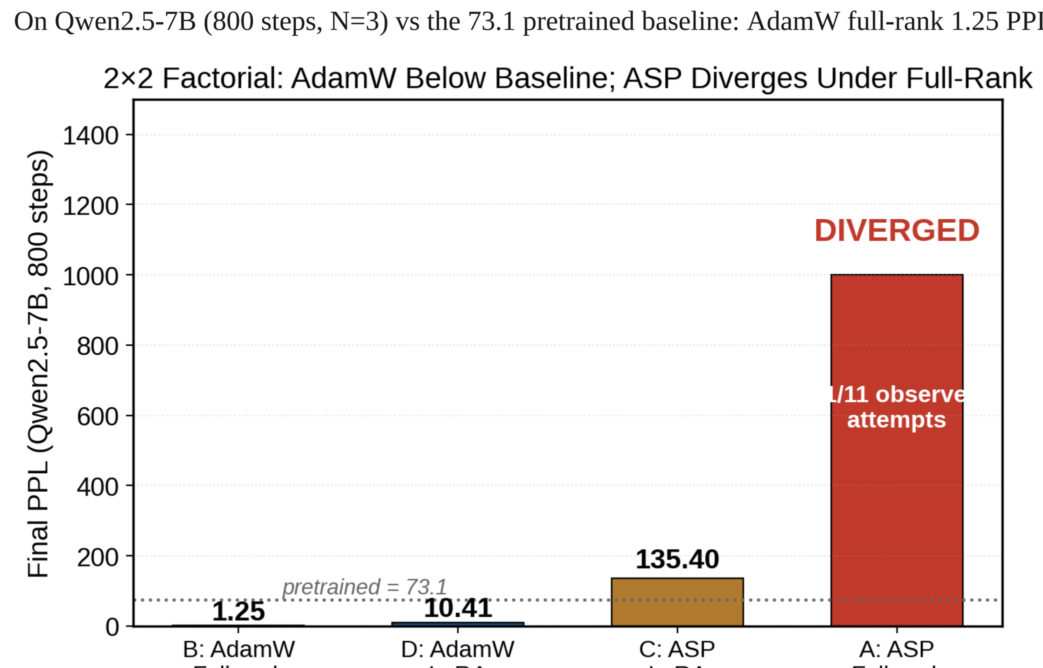
Finding 2 — Gradient injection: statistically indistinguishable from plain SGD

A-SYNC injects the ALS solution as a gradient bias; matched-budget pure SGD reproduces its trajectory (correlation 0.99981, mean per-cycle diff 0.116 PPL). No tested schedule beats plain SGD.



Finding 3 — The 2×2 matrix: AdamW below baseline, ASP diverges

On Qwen2.5-7B (800 steps, N=3) vs the 73.1 pretrained baseline: AdamW full-rank 1.25 PPL, AdamW+LoRA 10.41, ASP+LoRA 135.4, ASP full-rank diverges.



Verified Solutions (Qwen2.5-7B, 28 layers)

AdamW full-rank — 1.25 PPL (58× below baseline). Algorithm core: for every weight, compute the loss gradient (the slope that tells which way is downhill), then step down with an **adaptive per-weight step size** (Adam's moment estimates) plus weight decay. Repeated thousands of times, it descends the loss surface. It converges reliably at any depth and achieves the best absolute quality (N=3, 800 steps). *Use when*: quality is the priority and training compute is available.

AdamW + LoRA — 10.41 PPL (7× below baseline). Algorithm core: freeze the entire base model and train only a **tiny adapter ΔW=BA** (A: r×d, B: d×r, r ≪ d, ~0.1% of parameters) with the same AdamW updates. The adapter's low rank forces a compact, generalizable correction. Nearly the same FLOPs as full-rank (backprop through frozen weights dominates), but the optimizer state and adapter memory are orders of magnitude smaller. *Use when*: storage/memory is constrained or many task-adapters must be served.

Plain SGD, fixed lr — 6.83 PPL (10.7× below baseline). Algorithm core: the simplest descent — for every weight, take a small step opposite the gradient with a **single fixed step size (learning rate)**, never adapting it per-weight and never decaying it over training. It converges on 28 layers where ALS cannot — the decisive lesson that the failure lives in the ALS solve, not in gradient training. *Use when*: a simple, stable, dependency-free baseline is needed.

Method	Final PPL	vs 73.1 baseline	Best for
AdamW full-rank	1.25	**58× better**	absolute quality
Plain SGD (fixed lr)	6.83	**10.7× better**	simple stable baseline
AdamW + LoRA	10.41	**7× better**	memory / many adapters
ASP full-rank	DIVERGED	**11/11 attempts**	—

References & Reproducibility

[1] Hu, E. et al. LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022.
[2] Loshchilov, I., Hutter, F. Decoupled Weight Decay Regularization. ICLR 2019.
[3] Koren, Y. et al. Matrix Factorization Techniques for Recommender Systems. IEEE Computer 42(8), 2009.
[4] Merity, S. et al. Pointer Sentinel Mixture Models. ICLR 2017.
[5] Hinton, G. et al. Distilling the Knowledge in a Neural Network. arXiv:1503.02531.

All experiments reproducible: runs/ + experiments/ in the project repo. Evaluation-harness audit and per-step data archived. Scan to view the repository:

Project repository
github.com/gingersea/
alternating-optimization-lora

*Takeaway: on deep models, remove the ALS machinery and let gradient descent do its job with a sustained learning rate — **AdamW 58× better than the pretrained baseline, plain SGD 10.7×, LoRA 7×**. Best quality still requires AdamW.*

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